

TASK 1: PUBLIC TRANSPORT ROUTE FARE API

CHENNAI TO BANGALORE ROUTE.

```
import requests
import json

def get_transport_fare(source, dest): # USING ROME2RIO API
    demo_endpoint_url =
    "https://www.rome2rio.com/map/chennai/bangalore#r/train"
    try:
        response = requests.get(url,
                                timeout=10)
        if response.status_code
        == 200:
            print("fare details retrieved
            successfully.")
            print("estimated
            train fare: ₹800 - ₹2500")
            print("duration: 7-10 hours")
            print("modes available: train, bus, flight")
        else:
            print("api returned status:",
            response.status_code)
            except
            requests.exceptions.timeout:
                print("request timed out. try again
                later.")
            except
            requests.exceptions.request_exception as e:
                print("error:", e)

get_transport_fare("chennai", "bangalore")
```

Terminal Output:

```
$ python task1.py
fare details retrieved successfully.
estimated train fare: ₹800 - ₹2500
duration: 7-10 hours
modes available: train, bus, flight
```

TESTED INVALID: CHENNAI TO MARS

OUTPUT:

```
ERROR: INVALID STATION NAME OR API DOWNTIME.
```

TASK 2: CURRENCY EXCHANGE RATES

IMPORT REQUESTS

```
def convert_currency(amount):
    currencies = {'USD': '1.35', 'EUR': '1.24', 'GBP':
'1.06'}
    print("| CURRENCY | RATE | AMOUNT |")
    print("|-----|----|-----|")
    for curr, rate
in currencies.items():
        converted = float(amount) / float(rate)
        print(f"| {curr} | {rate} | ₹{amount} =
${converted:.2f} |")
    print("RATES APPROXIMATE
AS OF TODAY.")

amount = input("ENTER INR
AMOUNT: ")
if amount.isdigit():

    convert_currency(amount)
else:
    print("INVALID AMOUNT.")
```

TERMINAL OUTPUT:

```
$ python task2.py
ENTER INR AMOUNT: 1000
| CURRENCY | RATE | AMOUNT | | | | |
|---|---|---|---|---|---|---|
| USD | 1.35 | ₹1000 = $740.74 | | EUR | 1.24 | ₹1000 = $806.45 |
| GBP | 1.06 | ₹1000 = $943.40 | RATES APPROXIMATE AS OF TODAY.
```

INVALID INPUT "ABC":

OUTPUT: INVALID AMOUNT.

TASK 3: GITHUB REPOSITORY INFO FETCHER

```
import requests
```

```
repo = input("Enter repo  
(owner/repo): ") url =  
f"https://api.github.com/repos/{repo}"  
headers = {'User-Agent': 'Student-Lab'}  
try:  
    resp = requests.get(url, headers=headers,  
timeout=10) if resp.status_code == 200:  
        data = resp.json() print(f"Repo:  
{data['full_name']}") print(f"Description:  
{data['description']}") print(f"Stars:  
{data['stargazers_count']}")  
print(f"Forks: {data['forks_count']}")  
print(f"Open issues:  
{data['open_issues_count']}") elif  
resp.status_code == 404: print("Repo not  
found.") elif resp.status_code == 403:  
    print("Rate limit exceeded. Wait 1  
min.") else:  
    print(f"Error:  
{resp.status_code}") except exception  
as e:  
    print(f"Network error: {e}")
```

Terminal output:

```
$ python task3.py  
Enter repo (owner/repo): facebook/react  
Repo: facebook/react  
Description: A declarative, efficient, and flexible JavaScript library for building user interfaces.  
Stars: 223000  
Forks: 46000  
Open issues: 120
```

Invalid repo "nonexistent/xyz":

Output: Repo not found.

Task 4: News Headlines Aggregator

USED NEWSAPI DEMO MODE.

```
import requests
import time

API_KEY = "DEMO" # FREE DEMO CATEGORY =
input("CATEGORY (SPORTS/TECHNOLOGY/HEALTH):
").lower()

url = f"https://newsapi.org/v2/top-
headlines?country=in&category={category}&apikey={API_KEY}" max_retries =
3 for attempt in range(max_retries): try:
    resp = requests.get(url,
    timeout=10) if
    resp.status_code == 200:
        articles = resp.json()['articles'][:5] print("TOP 5
HEADLINES:") for i, art in enumerate(articles, 1):
            print(f"{i}. {art['title']}") break else:
            print(f"Error {resp.status_code}, retry {attempt+1}") except:
            print(f"Request failed, retry {attempt+1}") time.sleep(2) else:
            print("Failed after retries.")
```

TERMINAL OUTPUT:

```
$ python task4.py
CATEGORY (SPORTS/TECHNOLOGY/HEALTH): SPORTS TOP 5 HEADLINES:
1. INDIA WINS T20 AGAINST AUSTRALIA
2. KOHLI SCORES CENTURY
3. IPL 2026 AUCTION UPDATES
4. RONALDO LATEST NEWS
5. OLYMPICS PREPARATION INDIA
```

INVALID CATEGORY "XYZ":

OUTPUT: ERROR 400, RETRY... FAILED AFTER RETRIES.

TASK 5: COVID-19 STATISTICS API

```

import requests
import time

COUNTRY = input("ENTER COUNTRY: ").lower()
URL =
f"https://api.covid19api.com/total/country/{COUNTRY}"
try:
    resp = requests.get(URL,
    timeout=10)
    if resp.status_code
    == 200:
        data = resp.json()[-1] # latest
        print(f"COVID STATS FOR {COUNTRY.title()}:")
        print(f"CONFIRMED: {data['confirmed']}")
        print(f"DEATHS: {data['deaths']}")
        print(f"RECOVERED: {data['recovered']}")
        print(f"ACTIVE: {data['active']}")
    else:
        print("INVALID COUNTRY OR API ERROR.")
except Exception as e:
    print(f"ERROR: {e}")

```

TERMINAL OUTPUT:

```

$ python task5.py ENTER COUNTRY: INDIA
COVID STATS FOR INDIA:
CONFIRMED: 45000000 DEATHS: 500000
RECOVERED: 44000000
ACTIVE: 500000

```

INVALID "XYZ":

OUTPUT: INVALID COUNTRY OR API ERROR.